



Design and development of an indoor navigation system using denoising autoencoder based convolutional neural network for visually impaired people

J. Akilandeswari¹ · G. Jothi² · A. Naveenkumar¹ · R. S. Sabeenian³ · P. Iyyanar¹ · M. E. Paramasivam³

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Abstract

A challenging area of research is the development of a navigation system for visually impaired people in an indoor environment such as a railway station, commercial complex, educational institution, and airport. Identifying the current location of the users can be a difficult task for those with visual impairments. The entire selection of the navigation path depends upon the current location of the user. This work presents a detailed analysis of the recent user positioning techniques and methodologies on the indoor navigation system based on the parameters, such as techniques, cost, the feasibility of implementation, and limitations. This paper presents a denoising auto encoder based on the convolutional neural network (DAECNN) to identify the present location of the users. The proposed approach uses the de-noising autoencoder to reconstruct the noisy image and the convolution neural network (CNN) to classify the users' current position. The proposed method is compared with the existing deep learning approaches such as deep autoencoder, sparse autoencoder, CNN, multilayer perceptron, radial basis function neural network, and the performances are analyzed. The experimental findings indicate that the DAECNN methodology works better than the existing classification approaches.

Keywords Denoising autoencoder · Convolutional neural network · Indoor navigation · Classification · User positioning · Deep learning

1 Introduction

The number of visual disability individuals around the world is estimated at 253 million. Of these, 36 million are blind and 217 million are visually impaired [61]. 4.8% of the residents of this population are born with a visual disability (blindness), along with 90% of

✉ J. Akilandeswari
akilandeswari@sonatech.ac.in

¹ Dept. of IT, Sona College of Technology, Salem 636005, Tamil Nadu, India

² Dept. of BCA, Sona College of Arts and Science, Salem 636005, Tamil Nadu, India

³ Dept. of ECE, Sona College of Technology, Salem 636005, Tamil Nadu, India

the incidence of macular degeneration linked to age, diabetes, glaucoma, and injuries. If the world's elderly population grows exponentially, there is a projected increase in the incidence of vision loss associated with chronic disease [26]. The visually disabled person cannot travel from one position to another without any assistants. Researchers have been designing many blind navigation techniques/devices in recent decades. A commonly used navigation technology is the global positioning system (GPS). The closed area, such as houses, railway depot, airports, public offices, and hospitals, does not receive any GPS satellite signal. As a result, users would be unable to use the GPS service in such an environment [19]. Ultrasonic wave [44], radio frequency identification (RFID) [54], Bluetooth, WiFi [39], sensors [42] as well as smartphones [34, 62] recognize the users' present position. Some of the limits on indoor or predefined outdoor equipment/technologies are: (1) very expensive (2) heavy electricity consumption, (3) power loss (4) More entry points (5) high crash requirements. The choice of direction depends on the user's current position. The consumer position in an indoor navigation device is difficult to locate since users with visually impaired disabilities do not know the current location. Many methods such as computer vision, machine learning, and deep learning have just been used for the development of a navigation system. The approaches spontaneously allow visually disabled people to walk in the predetermined area.

Visual localization is a fundamental component of robotics, augmented and virtual reality, and some smartphone applications. In indoor and outdoor assisted navigation systems context, it plays a significant role. Visual localization is the process of searching a database for the best image that matches the query/test image. However, visual information for a site may vary significantly between query and database because of different environment fluctuations, which is difficult to achieve robust visual location. The process of visual localization is composed of two phases: (1) the offline phase, where the environment is acquired, and (2) the online phase, where the position is returned. Thoma et al. [52] formulated the map construction and self-localization problems using convex quadratic and second-order cone programs, respectively. In indoor and outdoor datasets such as COLDFreiburg and the Oxford Robotcar, the proposed technique is assessed where it is far superior to the previous approaches. In [28] the authors' proposed a visual localizer, which incorporates the features of ConvNet and global optimization, to achieve robust visual localization for assisted navigation for visually impaired people. This approach compares and analyses the performance of five major ConvNets including AlexNet, VGG16, GoogleNet, SqueezeNet, and MobileNet. From experimental data, the best performance for environmental invariance is thought to be offered by GoogLeNet. Variations between query and database images, lighting fluctuations, structural changes, dynamic objects, season variations, and viewpoint variations are all challenging issues in the visual localizer. As a result, visual changes in assisted navigation are more complicated. To overcome these constraints, the panoramic localizer is introduced. In a panoramic localizer, images are collected from huge Field of View (FoV) camera sequences, which encompass 3600 omnidirectional information about a location and can improve localization accuracy substantially. Fang et al. [12] presented a panoramic localizer that uses panoramas for omnidirectional perception and is based on coarse-to-fine descriptors. The proposed method is compatible with mobile navigation devices and may be used in several localization scenarios. In [8], a panoramic annular localizer is proposed, which includes a panoramic annular lens and strong deep image descriptors. The active deep descriptor is formed by processing and feeding the panoramic annular pictures acquired by the single-camera into the NetVLAD network, and sequential matching is used to create the localization result.

In this study, the visual localization method for visually impaired people's navigation is suggested, which combines the strength of de-noising autoencoder and CNN. In this navigation system, the current position of the users is determined using a deep learning-based classification algorithm. The training images are gathered from various angles in the following order; 0° for the west direction (backward), 90° for the north direction (left), 180° for the east direction (forward), and 270° for the south direction (right). Similarly, 30° and 60° for the northwest, 120° and 150° for the northeast, 210° and 240° for the southeast, and 300° and 330° for the southwest direction. These directions help to identify more features in the particular step and efficiently classify the query image. Dataset is also collected in different environments such as morning, noon, cloudy, and evening. Training the deep learning models on more data can result in more efficient models, and the augmentation techniques can create variations of the images that can improve the ability of the fit models to generalize what they are learning with new images. From the experimental results, it is believed that the proposed method efficiently classifies the images when compared to the existing approaches.

Deep learning is a type of machine learning that consists of multi-layered artificial neural networks. In pattern recognition, object detection, and classification, implementations use the process currently enabled by large data sets. The key goal is to develop a method of navigation for visually disabled people in indoor environments using deep learning approaches. The approach suggested determines effectively the current position of users and helps users to travel in the correct direction.

The approach suggested efficiently defines the users' current position and permits users to travel in the right direction. Autoencoder is a kind of neural feed-forward network where the input is compared to the output [41]. It encodes the input of the image into a smaller code and rebuilds the image output from it. It consists of three components: encoder, bottleneck, and decoder. It is a type of regularization. Autoencoders are used in many applications, including anomaly detection, dimensional reduction, compression of images, and de-noising of images. Autoencoder is classified into different forms, including autoencoders, de-noising, convolution, sparse, and vector. De-noising autoencoder is applied to rebuild the noisy image in the proposed solution. A convolutional neural network (CNN) is a deep learning algorithm that has more than one convolutional layer [25]. CNN's advantage over its predecessors is that it extracts the imperative features automatically, without any human intervention. This study combines the strength of de-noising autoencoder and CNN to effectively classify the current position of the users in the navigation path. Image augmentation techniques such as rotation, contrast, flip, brightness, zoom are applied in preprocessing to increase the size of the dataset [56]. This reduces over-fitting in the trained model. The deep neural network approaches like deep autoencoder [16], sparse autoencoder [33], Multi-layer perceptron (MLP) [51], and radial basis function neural networks (RBFNN) [47] are employed to assess the performance of the proposed approach. The trained model reduces overfitting. The following enumerates the research contributions:

1. The recent user positioning techniques and methodologies on the indoor navigation system are analyzed
2. Developed a navigation system for visually impaired people in an indoor environment.
3. Use de-noising autoencoder to reconstruct the noise-input image.
4. The convolutional neural network-based classification algorithm is employed to find the current location of the user in the navigation path

The proposed architecture is presented in Fig. 1.

The paper is organized as follows: Sect. 2 summarizes the relevant literature from this research and the different methods used to users position. Section 3 describes various deep learning neural network techniques. The proposed methodology is described in Sect. 4. The experimental findings are outlined in Sect. 5. Section 6 summarizes the conclusions and the future work.

2 Background and related work

In the past decades, numerous devices or techniques have been developed to provide support for visually impaired people and improve the quality of their lives by making them capable. In [32], the development of various assistive technologies and devices for visually impaired people is reviewed. They have also analyzed the important features of those devices and technologies with factors like power consumption, weight, economy, and user-friendliness. Dourado et al. [11] developed a navigation system for blind people using an evolutionary multi-objective optimization method. In this approach, morphological filters are employed to fix the missing distances in 3D images of RGB-D cameras. The developed method gives better performance when deployed in a real navigation system.

Mallick et al. [31] applied deep learning techniques to detect and classify the MRI brain tumor images. In this method, images are compressed using a deep wavelet autoencoder (DWA), and deep neural network (DNN) is utilized to classified the tumor and non-tumor images. The experimental results show that the proposed DWD-DNN approach performed better than the existing deep learning and non-deep learning classifiers such as Autoencoder-DBN, Multilayer perceptron neural network, DNN, Radial Basis Function Neural Network, Time Delay Neural Network, Probabilistic Neural Network. In [2] Hyperspectral imaging (HSI) is analyzed and has been extensively utilized in several fields. Generalized fuzziness extreme learning machine autoencoder-based active learning is proposed to classify the HSI. The proposed method reduces the sample selection bias whilst maintaining data stability in the spatial domain. In [27] the researchers investigated the Auto-Encoder model that helps to improve classification accuracy. In this approach, the latent compressed vector is generated by the encoder. The proposed

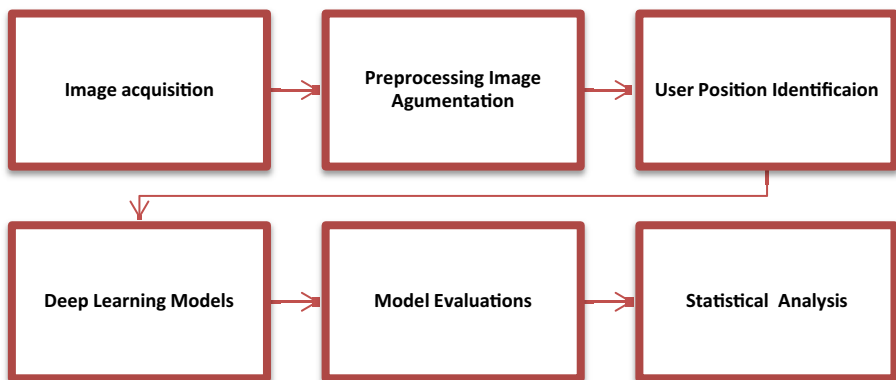


Fig. 1 The proposed framework

method is applied with the medical dataset of conjunctivitis and the STL-10 dataset and the performances were analyzed. The auto-encoder-based approach improved the classification accuracy and also helps to solve the problem of False Positive Rate.

Large margin auto-encoders (LMAE) was employed by Liu et al. [29] to build a deep neural network to learn the proper features. The performance of the LME approach proposed has been compared with the existing autoencoder. Two different datasets, such as MNIST and CIFAR-10, are used to assess classification performance and the results substantiated the benefits of the proposed approach. Karim et al. [21] have developed a new framework for medical waveform data sets, namely epilepsy, heart disease, and arrhythmias, based on deep autoencoder and energy spectral density. This method incorporates ESD as a feature extractor in the architecture of deep sparse auto-encoders (DSAEs), which helps with less computational time to extract more qualified features. The results show that, compared to existing researchers, the proposed architecture provides better results.

Zhang et al. [66] analyzed fourteen types of halftone images and classified high-quality continuous tone images. The features are extracted in this study using stacked Sparse Auto-Encoders (SAE). Softmax regression is used to fine-tune the deep neural network for the classification of images with halftone. The proposed approach to SAE-MV can differentiate the most categories of halftone images and attain competitive ACCR while showing better performance in generalization. Afif et al. [1] have used Deep Convolution Neural Networks as an indoor object detection technique (DCNN). This approach creates a new object dataset with a wide range of artifacts, indoor lighting, and indoor architectural frameworks which are potentially unsafe for VIP mobility. The experimental results show the proposed indoor object detection's high performance as its detection rate is 73.19% (average acute accuracy).

Gobi [15] has analyzed a new localization model with fingerprints using a bat algorithm (BA), which is stimulated by a microbat's echolocation. The proposed approach is used to improve the accuracy and to analyze the impact on the localization results of the initial location of the bats. Kalman filter is used for monitoring purposes to update the initially defined location using the BA. In the experimental study, real-time performance improvement and computational complexity decreases were suggested. The Neural Network Ensemble Learning for the indoor locomotive system is used for smartphone-dependent indoor location [45] (NNELILS). Here too the challenge of system heterogeneity is addressed. When the training and testing devices differ, the proposed method has improved the localization accuracy to 94%. In comparison to other indoor positioning methods, Shahid et al. [48] studied indoor localization using photos as an alternative. A cell phone camera is used to take photographs of the environment in the building to find the customer. There is no need for sensor data from different sensors or no additional system to locate the client in the building using an image-based indoor positioning. The client is located using Microsoft Azure Custom Vision Services. The average accuracy of the qualified model ranges from 90 to 95%.

Image-based localization using CNN and LSTM for structured feature correlation is proposed in [55]. In this method, a new TU Munich large-scale indoor (TUM-LSI) dataset is introduced for indoor localization, consisting of 1,095 high-resolution images covering a total area of 5,575m². Ground-truth posture information is connected with each image. From the experimental results, it is noted that the CNN-based (PoseNet) model performs better than the SIFT-based method i.e. it gives almost 30% in positional error and 55% orientation error. An overview of the related literature is shown in Table 1.

Table 1 Related literature

Author and year	Objective	Approach	Dataset	Evaluation metrics
Mallick et al. [31]	Detect and classify the mages of the MRI brain tumor	DWN DNN	MRI brain tumour images	Specificity Sensitivity F-score Accuracy Kappa Value
Li et al. [27]	Autoencoder is used to improve the accuracy of classification	Autoencoder CNN	Conjunctivitis STL-10	Accuracy
Liu et al. [29]	Large margin auto-encoders for classification models are investigated	Large margin auto-encoders autoencoder	MNIST CIFAR-10	Precision
Karim et al. [21]	Developed a new framework for medical waveform datasets based on the deep autoencoder and energy spectral density	Deep autoencoder Energy spectral density	Medical waveform datasets	Accuracy, sensitivity, specificity, F1 score matthews correlation coefficient
Zhang et al. [66]	Half-tone images are classified using sparse autoencoder and deep neural networks	DNN Sparse autoencoder	Half-tone images	Correct classification rate, average correct classification rate
Afif et al. [1]	To developed object detection techniques in an indoor environment	Deep convolutional neural networks	Object detection dataset	Mean average precision
Gobi [15]	Smartphone-based indoor localization and tracking model is developed using bat algorithm and Kalman filter	BAT algorithm	Indoor navigation dataset	Precision, classification accuracy
Roy [45]	To designed to smartphone-based indoor localization	Neural network-based ensemble learning	Indoor navigation dataset	Accuracy
Shahid et al. [48]	To developed an indoor localization model using a machine learning approach	Machine learning	Indoor navigation dataset	Average precision
Walch et al. [55]	To designed an image-based localization method for structured feature correlation	CNN LSTM SIFT	7Scenes dataset Cambridge Landmarks TUM-LSI	Localization accuracy

2.1 User positioning techniques

In this paper, we analyze the different user positioning approaches in an indoor environment. Figure 2 show the most commonly used methods of finding the actual location of the users in an indoor environment. The present location of the user is identified based on waves, sensors, and smartphones. Various algorithms are employed in each method such as location fingerprinting, location triangulation, and location trilateration. In brief, the sub-sections describe the algorithms for each of the categories.

2.2 User positioning based on the waves

The location of a person in a closed environment is generally done using sensors. These sensors utilize electromagnetic waves such as infrared, Bluetooth, Wi-Fi, ultrasonic waves, and light waves for identifying the location. Apostolos et al. [37], implemented the system using the Bluetooth beacons and dead reckoning algorithm to detect the location of the users in the navigation system. The Received Signal Strength Indicator (RSSI) is calculated from the beacon signals and the location of the users is found with the help of the dead reckoning algorithm. The author also developed the map matching concepts for indoor space. Xie et al. [63] presented a technique based on light intensity. On this, the angulations and distances of the light are calculated with the help of the Multi-Face Light Positioning (MFLP) principle.

The concept of MFLP is to detect the face of the lights using sensors and using that information with pre-defined RSS data, the location of the user is found. Do Trong-Hop et al. [10], the surveyed visible light-based positioning system. They analyzed algorithms used for finding the user location based on the light waves. Nakajima et al. [40], uses the light ID system with an external server to detect user positioning. Kirchner et al. [22] proposed the users' location identification technique based on infrared waves. The distance

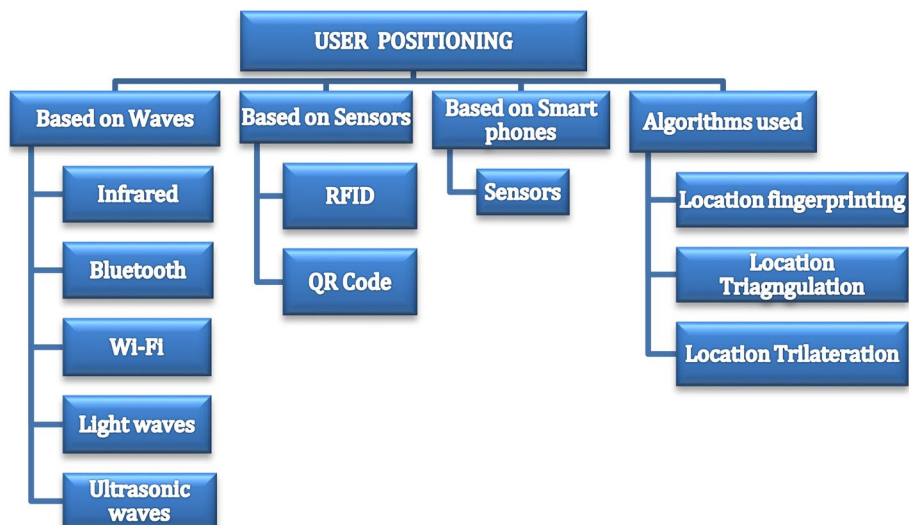


Fig. 2 Taxonomy of user position methods

Table 2 Waves and their ranges

Types of Waves	Range
Wi-Fi	Less than 150m
Bluetooth	Less than 75m
Ultrasonic	Less than 150m
Infrared	Less than 10m

between the transmitter and receiver of infrared signal and angles of a received signal is calculated and the user's location is found. Sakai et al. [46], developed the linear location-based system on IR beacon. In this method, the receiver is installed on the ceiling and assigned the ID for each receiver, and an IR beacon is attached with shopping bags. The ID of the receiver is used to locate the user. Amol et al. [43] presented the location position system based on the Wi-Fi signals. The combination of Trilateration, location fingerprinting methods, and RFID tags is utilized to predict the users' present location. Cardoso et al. [6], the user's location is identified using a Wi-Fi fingerprints access point which is mounted in the wall. The different types of waves and their ranges are presented in Table 2.

The literature reviews of techniques for user position identification based on electromagnetic waves are summarized in Table 3.

2.3 User positioning based on the sensors

Kushagra et al. [50], proposed a user location identification system using Radio Frequency Identification [RFID] tags with GPS. The location information is fetched from the RFID tags and the reader is used to retrieve the information. Ivanov [18], the RFID tags are employed to locate the user. In this method, the retrieved information is stored in the WAP binary eXtensible Markup Language (XML) format. This format helps to increase the effective use of XML data on narrowband communication channels. Kassim et al. [23] introduced the digital compass with RFID tags to identify the users' position and localization. Willis et al. [60] implemented decentralized methods to detect the user location using RFID. In this approach, the RFID grid information is fetched with the help of service robotics.

Carlos et al. [36], introducing the TELIAMADE system for navigation system based on the time of flight (TOF) of ultrasonic waves. The proposed system is used to positioning the user with calculating the time of flight between the receiver and transmitter node. The ultrasonic channel is managed by applying a time-division multiplexing scheme. Jun et al. [20] use the least square method to calculate the time of flight ultrasonic waves between the source and receiving node. User location is calculated based on the least square method. Ugar et al. [65] calculated the user position by fixing the ultrasonic transmitter at known places and using the time difference arrival method to locate the user.

Farshad et al. [13] introduced the QR-based user location for navigation purposes. In this system, three big squares in the corners (finder pattern) were used for detecting the

Table 3 Summary of waves based methods for user position identification

Authors and year	Techniques	Objective	Cost	Feasibility level	Demerits
Apostolos et al. [37]	Bluetooth waves	Developing a blind museum tourer application to navigate the blind people on the indoor location of the museum	High (Bluetooth beacon)	Complex (More calculations are needed in dead reckoning algorithm)	User positioning fully relies on Bluetooth beacons and proximity sensors
Xie et al. [63]	Visible lightwaves	Designing the LIPS receiver (CompEye) for locating the user position using three sensors such as light-sensing component, magnetic sensors, and accelerometer sensors	High (Microcontroller and sensors)	Complex (Detecting the light faces takes complicated calculations)	CompEye receiver consumes more power and LED lights must be labeled
Nakajima et al. [40]	Visible lightwaves	Implementing the indoor navigation system for visually impaired people using LED with geomagnetism	Moderate (Wireless LAN a for creating MAP)	Moderate (Creation of indoor map data)	Not suitable for the real-time environment
Kirchner et al. [22]	Infrared waves	Implementing IR-LPS for navigating the UAV on a non-calibrated environment using a single IR transmitter and Three IR sensing sensors	High (Designing the Board for sensing the IR signal)	Moderate (Triangulation methods only involve relies on fewer calculations for user positioning)	Difficulties of hardware design
Sakai et al. [46]	Infrared waves	Proposing the IR beacon-based navigation and also designed the array of a receiver with angle detection	High (Array of a receiver with angle detection)	Moderate (Conversion of IR waves into Analog waves)	Design of receiver with angle detection
Amo et al. [43]	Wi-Fi waves	Implementing the navigation system by combining RFID tags and Wi-Fi signals	High (RFID tags, Wi-Fi Access Point)	High (Triangulation and fingerprint concept depends on the calculation of received signal strength)	Infrastructure cost is high
Cardoso et al. [6]	Wi-Fi waves	Developing the navigation system (w4all) for university campus based on Wi-Fi	Moderate (Wi-Fi access point)	High (Calculation of each access point fingerprint)	Infrastructure cost is high

user position. Ilkovicova et al. [17] proposed the QR-based positioning system. The location information is stored in the QR code. The user positioning information is calculated by scanning the QR code. José et al. [36], proposed the method that combines the QR code and NFC to locate the user. The geographic information is perfected on QR codes and NFC chips. While scanning the QR code or NFC chips the user location information is found. Table 4 described a summarization of related work for sensor-based models.

2.4 User positioning based on the smartphones

The development of the smartphone industry can connect the internet anywhere and anytime. This leads to lots of development on location-based systems. Wang et al. [57] introduced the machine learning algorithm to positioning the user by classifying the signal strength received from neighbour GSM base station. Waqar et al. [58], uses the information received by the gyroscope, accelerometer, and electronic compass sensors of smartphones to locating the users without the need for additional infrastructure. Martin et al. [59], the user locations are estimated with the help of an image recognition system and distance estimation algorithm.

In the smartphone and its embedded sensors including Wi-Fi, Bluetooth, accelerometer, and a gyroscope, Mashuk et al. [35] developed a multi-floor indoor positioning system. This article also analyses indoor positioning system real-time implementations for occupancy detection in the environment of buildings. In [62] an indoor system with smartphone cameras was planned. Static artifacts such as doors and windows are known as reference points in this model. The smartphone cameras captured the reference point and then find the user's location. Table 5 offers a summary of the literary study of strategies of user identification based on smartphones.

2.5 Algorithms based user positioning

Sensing the location of the user is achieved using various techniques such as electromagnetic waves, sensors, and smartphones. Despite these techniques, some algorithms are utilized to find the users' position based on Received Signal Strength (RSS) and classified into three different methods such as fingerprinting, trilateration, and triangulation. Location fingerprinting is a way to find the location within the wireless network. In this method, the signal strength values are observed during the offline method. These values are compared with the real-time (online) observed signal strength and the location is sensed. Taheri et al. [49] developed the locus framework to identify the location based on fingerprinting. Bonde et al. [5] introduced the program to calculate the location based on fingerprinting.

Location trilateration is an alternative method for finding the location of the user. This method maintains the distance data based on the received signal strength percentage instead of maintaining offline signal strength. Mahiddin et al. [30], worked based on the trilateration method. Wi-Fi analyzer of a smartphone is used to calculate the percentage of received signal strength from the access point. Chan et al. [7], developed a location-based service system based on both fingerprinting and trilateration concepts. Location triangulation is an extension of the Trilateration algorithm. In the triangulation method, angles of signal from the transmitter are calculated. Triangulation determines the distance by forming the triangles based on the angle value. Cieczkowski [9] introduced the static IR beacon-based triangulation for user positioning.

Table 4 Summary of sensors based methods for user position identification

Authors' and year	Techniques	Objective	Cost	Feasibility level	Demerits
Kushagra et al. [50]	RFID sensors	Developing the electronic oriented aids based navigation system in a building with the help of an embedded RFID reader in the stick and providing voice assistance	High (Blind man stick)	Difficult (Complex design of blind man stick)	The power consumption of the RFID reader is high
Ivanov [18]	RFID sensors	Proposing the java program to read the RFID tags for an indoor navigation system	Moderate (Two RFID tags are used. One is used for Navigation purpose and another one is for audio purpose) High (RFID)	Moderate (Storing navigation information in XML format and audio in adaptive multi-rate format)	Navigation and obstacle detection fully relies on an audio tag and white cane
Kassim et al. [23]	RFID sensors	Implementing the localization and positioning rest on the embedded digital compass and RFID tags for indoor navigation	High (RFID)	Difficult (Complicated design of embedded white cane)	No proper method to detect the obstacle or object
Willis et al. [60]	RFID sensors	Recommending the idea of storing spatial coordinates on RFID tags (RFID grid information) for indoor navigation system without relies on a central database	High (Proximity sensor, RFID reader embedding with Cane and Laying the RFID in the Carpet)	Moderate (spatial coordinates are stored in the form of XML)	Power consumption of RFID tags and obstacle detection
Carlos et al. [36]	Ultrasonic sensor	Introducing the TELAMADE system for navigation system based on the TOF of ultrasonic waves	High (Amplifier, transducer, Analog to Digital converter)	High (Hardware design is hard and TOF calculation is difficult)	Difficult to locate the user from a different angle
Jun Qi et al. [20]	Ultrasonic sensor	Proposing the new time-domain method to estimate the TOF in the ultrasonic indoor positioning system (UIPS)	High (Mobile robot, sensors, ultrasonic transmitter, filter)	Moderate (Estimation of Time and TOF)	Need more number of anchor nodes to detect the ultrasonic waves

Table 4 (continued)

Authors' and year	Techniques	Objective	Cost	Feasibility level	Demerits
Ugar et al. [65]	Ultrasonic sensor	Presenting the ICKON system for indoor navigation system by estimating the time difference of arrival between the mobile node and ultrasonic transmitter	High (MKB, HKB, and GAB)	High (Estimation of TDOA)	Infrastructure cost is high
Farshad et al. [13]	QR code	Proposing the quick response (QR) technology for location-based services	High (Data provider, service provider)	Moderate (Integrating the data provider, service provider, and mobile phone)	This system fully relies on the third party such as data provider and service provider
Ilkovičová et al. [17]	QR code	Developing the android application for indoor navigation by using the QR code to positioning the users	Moderate	Moderate	Working on a centralized database

Table 5 Summary of smartphone-based methods for user position identification

Authors'	Techniques	Objective	Cost	Feasibility level	Demerits
Wang et al. [57]	Machine learning	Machine learning algorithms are used to classify the GSM base station signals to find the user location	High	High (To calculate the signal strength)	Loss of signal strength in the indoor environment because it is worked based on the GPS and GSM signal
Mashuk et al. [35]	Smartphone sensors	Designed a multi-floor indoor positioning system using smartphone sensors	High	Difficult (To create an interface for monitoring all the sensors of smartphones)	Need more number access point for more accurate results and also more collision
Xiao et al. [62]	Smartphone camera	Static objects are identified by a smartphone camera and find the user location	Low (Does not need any extra infrastructure)	Difficult (Similar static objects presented)	It is very difficult to find the location when the obstacles are presented in front of static objects

3 Deep convolutional neural network techniques

3.1 Multilayer perceptron (MLP)

Each neuron in MLP neural network takes the weighted sum of its input values. In other words, an equation multiplies each input value, and the results are all added up together. A single MLP neuron is a simple linear classifier, but the integration of such neurons into a network can produce complex non-linear classifiers [14]. The architecture of the MLP is shown in Fig. 3.

Mathematically, a neuron can be defined as follows [5]:

$$y = f(l + b) \quad (1)$$

$$l = \sum_{i=1}^n x_i w_i \quad (2)$$

The input vectors are denoted as $x_1, x_2, x_3, \dots, x_n$, the weights are denoted as $w_1, w_2, \dots, w_n$, l is the linear combination of the weighted inputs, b is the bias value, f is the activation function, and y is the output vector. The layers are completely connected in multilayers which means that each unit in a layer is connected to all units in the previous layer which have a specific set of weights.

3.2 Radial basis function neural network

Radial base function neural network (RBFNN) is the type of neural network that solves problems of classification and regression [38, 47]. This is an elegant algorithm for machine learning consisting of three distinct layers. The first layer is an input layer, the second layer is a hidden layer consisting of several non-linear RBF activation units, and the last one corresponds to the network's final output. Activation functions in RBFNs are implemented conventionally as Gaussian functions. The architecture of RBFNN is given in Fig. 4.

The input layer consists of an n -dimensional vector that displays to every neuron of the RBF. Every RBF neuron stores a prototype vector from the training set in the hidden layer. In the figure, it is noted that the RBF neuron is represented in a bell curve and the central value is called the center of the prototype vector or neuron. Comparing the input vector to its reference, it measures the resemblance based on Gaussian Equation (Eq. 3).

Fig. 3 Architecture of the MLP

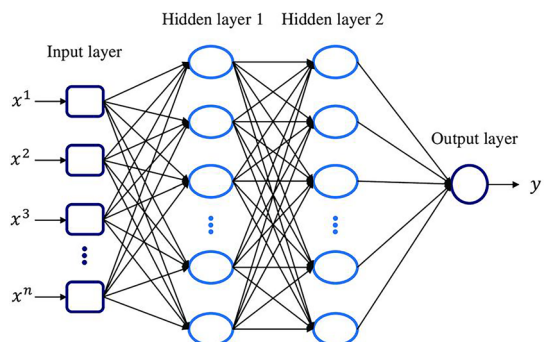
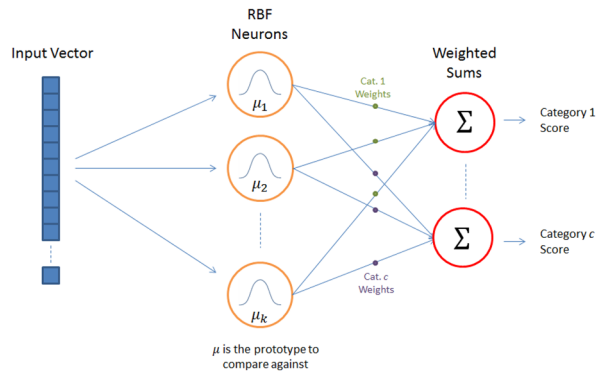


Fig. 4 The RBFNN architecture

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \quad (3)$$

where x is the input vector, the mean is μ and the standard deviation is σ . If the value of similarity is 1 the input is equal to that of the prototype otherwise it will give 0. The activation function of RBF neurons is slightly different, and is usually written as:

$$\varphi(x) = e^{-\beta\|x-\mu\|^2} \quad (4)$$

In the Gaussian distribution, μ corresponds to the distribution mean. It refers to the center of the neuron which is in the middle of the bell curve.

3.3 Convolutional neural network (CNN)

Convolutional Neural Network (CNN) is a subset of deep learning neural networks [25]. It represents a huge advance in image classification and recognition. CNN contains a series of layers and each layer in ConvNet is combined with activation functions. There are four major layer forms in CNN's architecture: the input layer, the convolutional layer, the pooling layer, and the fully connected layer. The convolution operation derives the low- to high-level features from the input layer in a convolutional layer series. The pooling layer limits the space size of the Convolved features. Two kinds of pools exist Max and average pools. There are two types. Max Pooling returns the full value and Average Pooling represents the average of all data points in the kernel region. Fully connected layer measures and classifies class label value through a way of softmax classification. The architecture of CNN is shown in Fig. 5.

4 The denoising auto encoder based convolutional neural network (DAECNN)

4.1 Denoising autoencoder

An autoencoder is a kind of neural network approach used for data denoising, reduce dimensionality and compress data. Denoising is a simple autoencoder extension [64]. The

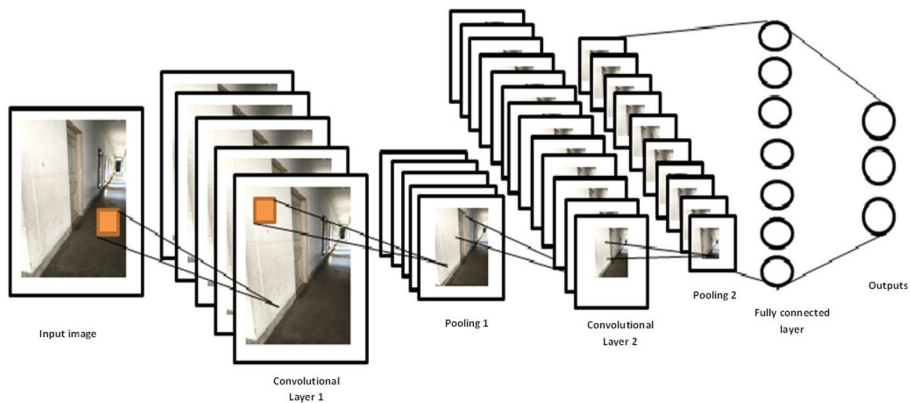


Fig. 5 The CNN architecture

denoising autoencoder knows how to efficiently compress corrupted data (encoder) and then restore encrypted data that is similar to original data (decoder).

4.2 The DAECNN classification algorithm

In this research, an efficient navigation system was built for visually impaired persons in a closed environment, using two separate neural networks, denoising autoencoder, and convolution neural network. DAECNN Architecture as seen in Fig. 6.

Denoising auto encoder (DAE) is applied in the proposed approach to reconstruct the original image from the corrupted input. The classification algorithm for the CNN is used to categorize the location of the user from the reconstructed image. Let the image pixel set $X = \{x_1, x_2, \dots, x_n\}$ and the output image $Y = \{y_1, y_2, \dots, y_n\}$.

Adding noise is a kind of regularization. Therefore, noise is to be added to the image input and referred to as $\tilde{X}$. The DAE uses the following Eq. (5) to map an input vector $\tilde{X}$ to a hidden representation called as an encoder $e(\tilde{X})$ [64].

$$h = e(\tilde{X}) = \varnothing(b + W\tilde{X}) \quad (5)$$

where W is a $m' \times m$ weight matrix, b is a bias vector and $\varnothing$ is a function of sigmoid activation. A reconstructed vector $d(\hat{X})$ maps the hidden representation (h) sometimes called the bottleneck or latent representation.,

$$d(\hat{X}) = \varnothing(b' + W'h) \quad (6)$$

The loss function $l(f(X))$ is defined as follows,

$$l(f(X)) = \frac{1}{2} \sum_i (\hat{X}_i - X_i)^2 \quad (7)$$

A linear activation function is used to reconstruct the loss, i.e., the sum of square differences. The output image is compared with the original image in the loss function and not a noise image. It shows how effectively the trained model reconstitutes the original image.

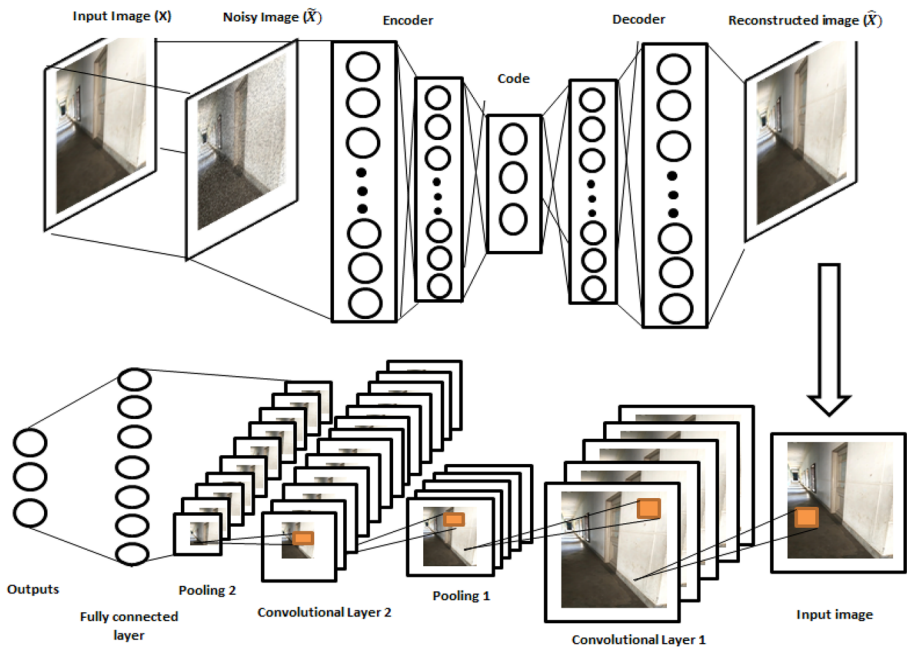


Fig. 6 The DAECNN architecture

The image reconstructed is then used as a model for the convolution neural network (CNN). A stack of convolutional layers accompanied by bundling layers that help prevent overfitting is used in the CNN model. Finally, a dense layer accompanied by a softmax layer is used. The convolution layer is often called a feature extractor layer because filters in this layer extract the image features. It also requires ReLU activation to zero all negative values. ReLU feature activation, as defined below [64],

$$f(x) = \begin{cases} x, & \text{where } x > 0 \\ 0, & \text{Otherwise} \end{cases} \quad (8)$$

Let $M \times M$ be the image size and $F \times F$ be filter size after convolution layer the output matrix size will be

$$(M \times M) \times (F \times F) = (N - F + 1) \times (N - F + 1) \quad (9)$$

Convolution stride is another control variable that defines how many steps in convolution we are moving at each step. This operation only used the edges of the image once in this process and rapidly reduced the dimensions of the input image. That results in information loss. To avoid that loss padding is used. Padding is an insertion of zeros in the input matrix by symmetrical methods. It is used to render the output image proportions as the input image. Padding is included in the input matrix $(M + 2p) \times (M + 2p)$ and the output matrix dimension is obtained as follows,

$$(M + 2p) \times (M + 2p) = (M + 2p - F + 1) \times (M + 2p - F + 1) \quad (10)$$

After the conversion, the pooling layer is used to reduce the input image volume. It is used among two convolutionary layers. A fully connected layer is used to distinguish images by using a Softmax activation to connect the neurons in one layer to the neurons in another.

5 Experimental analysis

5.1 Experimental setup

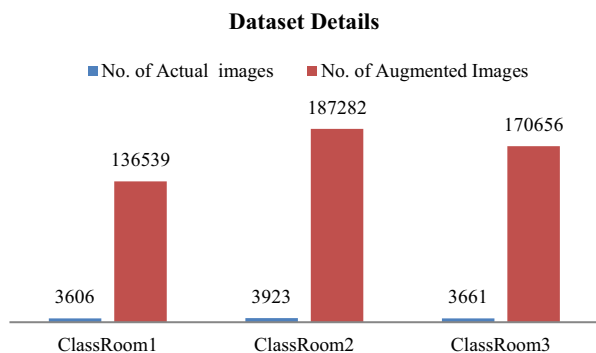
The experiment was executed in NVIDIA tesla v100 NVlink 32 GB 2GPU, Intel Xeon processor (12 core) with 128 GB RAM. The proposed approach is implemented with Python in Keras and TensorFlow.

5.2 Dataset and preprocessing

For this experimental analysis, 11,190 images are collected with three different classes like Class Room1, Class Room2, and Class Room 3 in the navigation area. 360° videos are taken using the Android phone Honor 8 Pro with the pixel resolution of 2976×3968 and then image frames are generated from the videos. The dataset is divided into three separate datasets, the training, validation, and test datasets (80% of the images as training, 10% of the image as validation, and 10% of the images as test data). Image augmentation parameters are used to artificially expand the size of the dataset. The parameters for this are zoom, shear, rotation, contrast, flip, brightness, etc. and 4,94,477 images are obtained for training after the augmentation. Figure 7 displays the number of actual images and the augmented ones. For each class, i.e., Classroom1, Classroom2, and Classroom3 the total number of samples increased by 38, 48, and 47%, respectively. This improves the training accuracy of the proposed neural network model.

To train the model, the preprocessed images are fed into the input layer. The inputs and the augmented images are shown in Fig. 8.

Fig. 7 Actual versus augmented samples



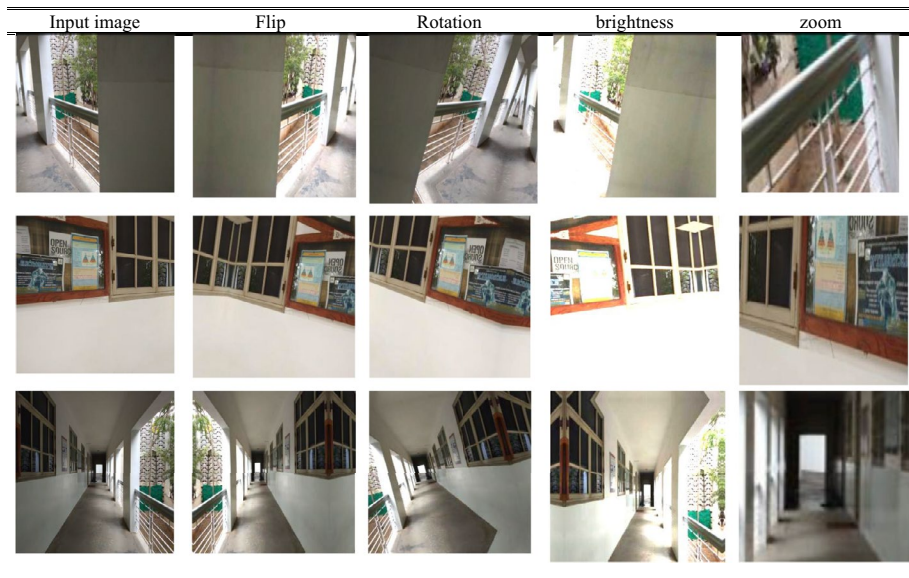


Fig. 8 Augmented images

5.3 Pilot user study

In the experimental analysis, a survey has been conducted with visually impaired people among 50 visually impaired people of the National Institute for the Empowerment of Persons with Visual Disabilities (Divyangjan), Chennai, and the Indian Association for Blind (IAB), Madurai. The following are the collected data related to navigation:

1. How do you want navigation instructions whether in the form of audio or text?
2. Do you need instructions for every step including turns?
3. How to represent the objects in the navigation path?
4. Do you need the number of steps in the staircases?
5. How they are using smartphones?

Based on the collected data, the deep learning model and mobile application has been developed. After completion of this application, a subject test has conducted on the developed navigation with 10 visually challenged people. The summarization of the subject testing feedback is presented in Table 6.

6 Results and discussion

In this study de-noising, autoencoder-based convolutional neural network (DAECNN) algorithm is employed to classify the current position of the users. The training dataset consists of three target classes (user positions) namely, classroom1, classroom2, and classroom3. The denoising autoencoder is applied to reconstruct the image from the corrupted image. The input image is accompanied by Gaussian noise. This process helps to improve

Table 6 The pilot user study

Suggestions	Action taken
Change the instructions like, go straight, turn left or right. Don't give the steps count every time	Constraints are added
Inform the objects/obstacles approximate distance	To consider for future development
Inform the staircase information before two or three steps in the navigation place	Corrected
It is difficult to find the position of the mobile camera	Give the pouch for fitting the mobile camera in the standard position
The Espeak voice assistant has to be integrated with the navigation application	To consider for future development
The application to incorporate the nearby person identification	To consider for future development

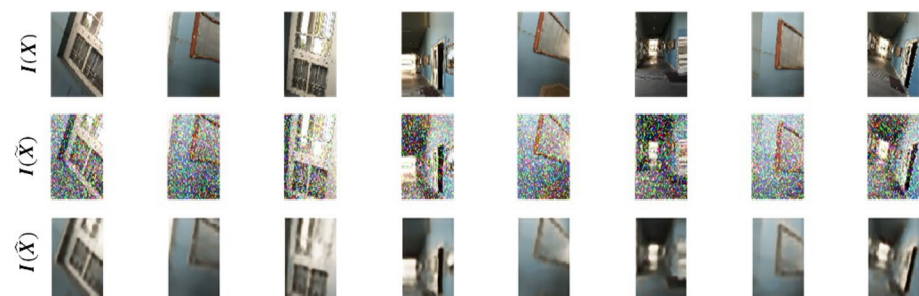
the performance of the CNN model to effectively classify the current position of the users. The results of the DAE approach are presented in Table 7 and the first, second, and third rows represent the original image $I(X)$, noisy image $I(\tilde{X})$, and reconstructed image $I(\hat{X})$ respectively.

The fully convolution layers in the CNN model learn the features i.e., edges, shapes, colours, angles, etc., and assigns a combination of features to a specific output. In the learning process the image dimensionality is reduced (Encoder and bottleneck). This 'compressed state' is known to reflect the latent space of an image shown in Fig. 9. This eliminates the 'extraneous' information that is distinct from each image and preserving the most relevant features.

The key advantage of Latent space representations is transforming more complex forms of raw data, i.e. images or videos, into simpler representations that are more convenient for processing and analyzing. In this proposed approach, the encoder works by mapping images from a dimension space of 150,528 ($224 \times 224 \times 3$) to a dimension space of 2048 (latent image). The decoder works by mapping back the latent image into dimension space of 12,288 and the results are shown in Fig. 10.

The reconstructed image is then fed as input to the CNN classification model after the noise has been removed. The feature maps extracted from the convolution layer are visualized in Fig. 11.

The parameter utilized in this study is presented in Table 8.

Table 7 Results of DAE approach

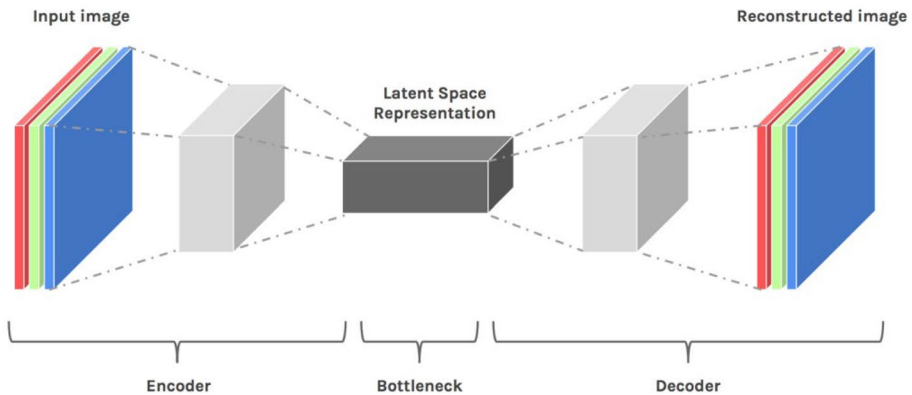


Fig. 9 Representation of convolutional neural network. Source: Hackernoon Latent Space Visualization

Convolution layers with 32 filters/kernels with a window size of 3×3 are added to implement the CNN model. The activation function 'Relu' is computed in convolution operation. A convolution layer with 64 filters followed by the max-pooling layer with window size 2×2 and a convolution layer with 128 filters can be added similarly. A drop-out layer with a 0.25 ratio is then added. The flattening layer transforms the three-dimensional array into a one-dimensional size array. Finally, the dense layer uses a softmax layer to classify the labels. CNN classification model is then utilized to identify the user's current position in a predefined environment. The de-noising auto-encoder-based CNN method is used to make predictions on the test images. Figure 12 shows the DAECNN model performance for the correct classification of test images.

Figure 13 indicates the test image dataset's misclassified groups. Since these images have the most similar features.

6.1 Ablation study

In this research, designed the CNN model has ten hidden layers and one dense layer. And softmax is used as an activation function. 11,190 images are collected with three different classes like Class Room 1, Class Room 2, and Class Room 3 in the navigation area. Images are taken using the Android phone Honor 8 Pro with a pixel resolution of 2976×3968 . The dataset is divided into three separate datasets, the training, validation, and test datasets (80% of the images as training, 10% of the image as validation, and 10% of the images as test data).

Initially, the model has trained on the GPU server (2Nos NVidia V100-32 GB), Intel Xeon processor (12 core) with 128 GB RAM. On this training phase dropout has set to 0.5 with epoch 500 and obtained the final test accuracy of 84.85%. The deep learning model has been converted into a TFlite model to integrate with the android application. The developed application is tested on the navigation path. The prediction's accuracy is insufficient. As a result, image augmentation is applied to the dataset in order to increase the model's accuracy. After performing the augmentation, the number of images in the dataset is around 4,94,477. After changing the above things on the model, again model has trained with the same epoch on the server. This time the final test accuracy of the model is 87.51%. Again the newly trained model is tested on the navigation path. At

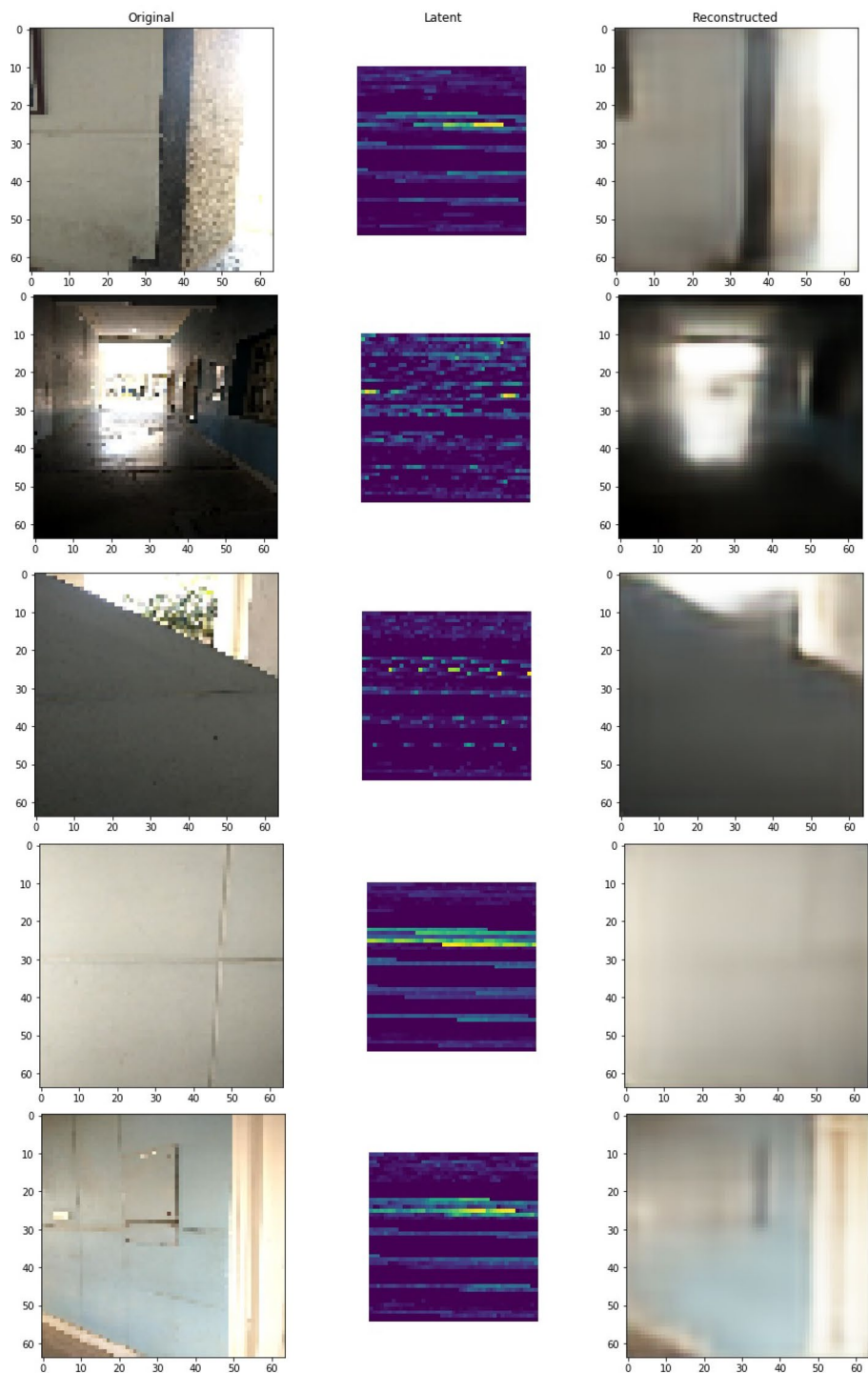


Fig. 10 Latent space representation

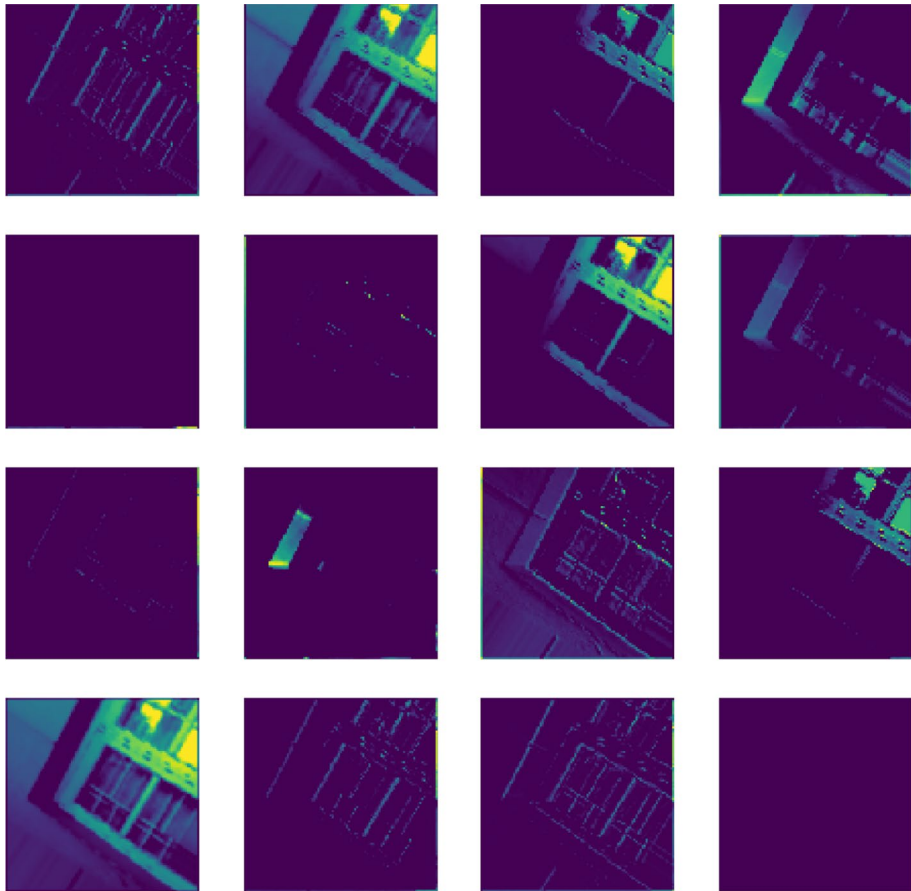


Fig. 11 Visualization of the feature maps

this time, the navigation app prediction has increased. But it is not sufficient to navigate in the real-world environment. Again the model is retrained to improve the accuracy of the model with activation function softmax. At this time, batch normalization has been included in the training phase along with dropout (0.25) and obtained the test accuracy of 94.37%. Finally, the model is integrated with a denoising autoencoder to remove the noise in the dataset and the CNN model has retrained with the new dataset. The final test accuracy of 94.37% has been observed. The model was then retrained with half of the hidden layers. However, removing or half the hidden layers leads to degradation in the final accuracy.

From this study, the following components of the proposed networks are identified as important.

- Dataset size
- Dropout value
- Batch normalization
- Denoising encoder

Table 8 Parameter utilized

Parameter utilized	MLP	CNN Model	DAE	RBNFNN	SAE	DAECNN
Input image size	$224 \times 224 \times 3$	$224 \times 224 \times 3$	$224 \times 224 \times 3$	$224 \times 224 \times 3$	$224 \times 224 \times 3$	$224 \times 224 \times 3$
Filter sizes	NA	[32, 64, 128]	[32, 64, 128]	NA	[32, 64, 128]	[32, 64, 128]
No. of hidden layers	15	15	25	15	20	10
Number of units in fully connected layer	20	30	30	30	30	30
Dropout rate	0.25	0.25	0.3	0.25	0.25	0.25
Learning rate	0.001	0.001	0.001	0.001	0.001	0.001
Number of epochs	500	500	500	500	500	500
Batch size	50	50	50	50	50	50
Activation function	RELU	Softmax	RELU	RELU	RELU	Softmax

6.2 Computational efficiency

Generally, the performance of the navigation application is measured by the inference time and memory space. The performance of the developed android navigation application is measured by using a System tracing app. The system tracing app works based on the

**Fig. 12** Correctly classified results

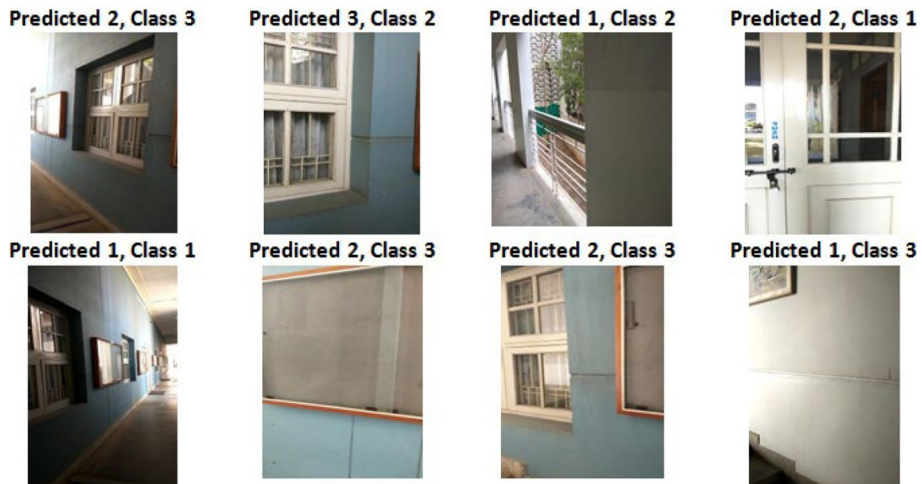


Fig. 13 Misclassified results

android benchmark. This app captures the TFlite events and measures the inference time of the prediction. The system tracing app is used on the Redmi Mi10i mobile device to measure the performance of the navigation application. According to the results of the system tracing app, the developed application predicts the class at 15.31 ms. Figure 14 shows the inference time of the developed navigation app. This app takes an installation memory of 38.9 MB.

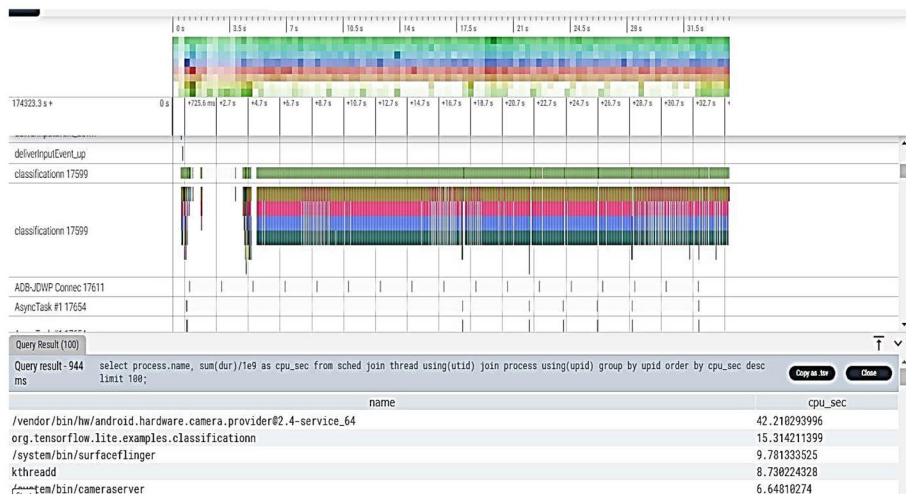
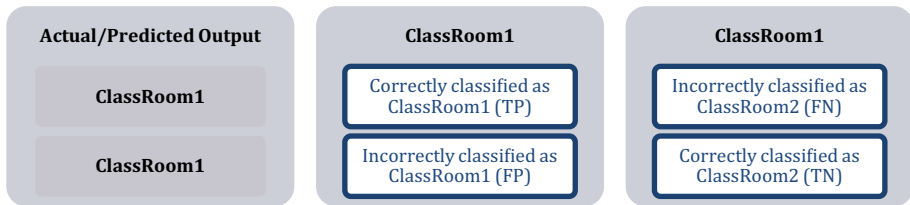


Fig. 14 Inference time computation

Table 9 Confusion matrix

6.3 Performance analysis

In this experimental study, we have explored the following classification approaches: Deep autoencoder (DAE), Sparse autoencoder (SAE), Multi-Layer Perceptron (MLP), CNN, Radial Basic Neural Network (RBFN), and DAECNN algorithms to identify the user location within a closed environment. These techniques are employed for evaluating the DAECNN algorithm's output. Different assessment measures such as accuracy, precision, recall, F1-measure, and error rate are used to test the performance of classification approaches. The equation of each evaluation measure is shown as follows [3, 4, 53],

$$\text{Precision} = TP / (TP + FP) \quad (11)$$

$$\text{Recall} = TP / (TP + FN) \quad (12)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (13)$$

$$F1_Score = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (14)$$

$$\text{Error_rate} = FP + FN / TP + TN + FP + FN \quad (15)$$

A confusion matrix displays the amount of correct and incorrect estimates of classification methods relative to the actual findings of the data. The sample confusion matrix is shown in Table 9.

Table 10 summarizes the classification findings from various classification algorithms in terms of performance, recall, and F1 measure, which were considered in this research. The experimental findings suggest that the DAECNN method performs better than the existing deep learning classification methods. Looking more carefully at the average

Table 10 Classification results

Classification approach	Precision	Recall	F1_score
DAE	77.49	81.53	82.67
SAE	75.27	71.62	76.35
MLP	71.64	78.07	72.26
RBNFNN	85.17	88.47	89.12
CNN	90.31	90.14	91.01
DAECNN	94.37	94.67	93.10

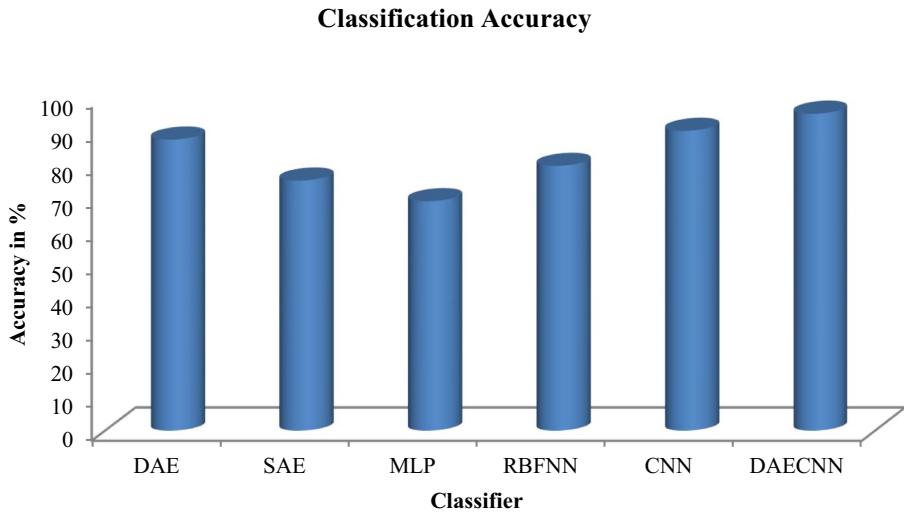


Fig. 15 Classification accuracy

results of the DAECNN method the strategy is thought to deliver the most reliable i.e. 95.37%. The MLP approach also results in the lowest accuracy of 67.20%.

The classification accuracy of each of the traditional approaches is shown in Fig. 15. In this figure, it is noted that the proposed Denoising autoencoder-based CNN (DAECNN) approach performs better than the existing approaches.

The error rate is determined by calculating the number of classified, unlabeled images by the total image number. The range of error rate lies between zero and one, however, the best model gets 0.0 and the worst one is 1.0. Figure 16 displays the error classification values for various classifiers. The DAECNN approach suggested has the highest error rate, i.e. less than 0.03.

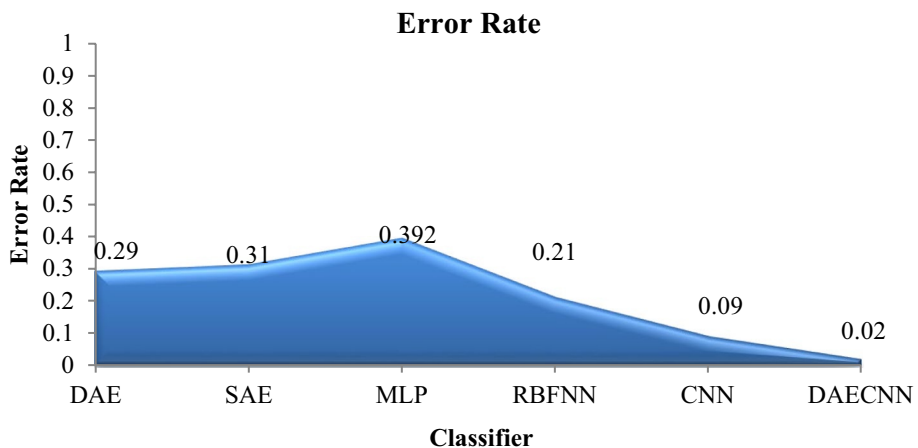


Fig. 16 Error rate

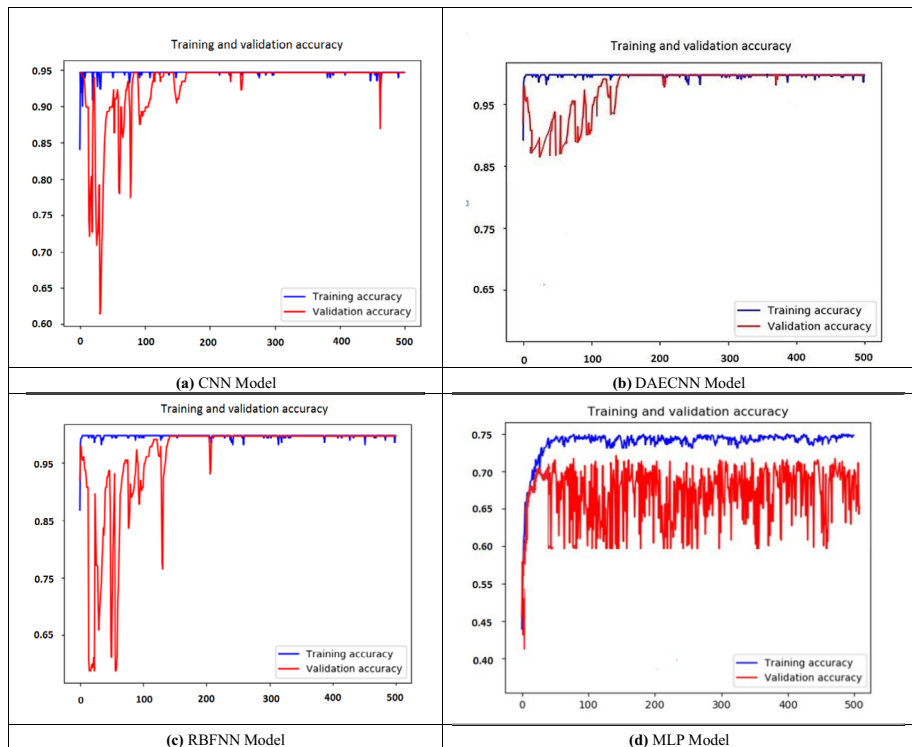


Fig. 17 Training and validation accuracy

In Fig. 17, the training and validation accuracy for CNN, RBFNN, MLP, and DAECNN is presented. The training and validation accuracy depends on the number of epochs which is denoted as blue and red color respectively. The validation accuracy was measured using a 10% additional dataset of validation dataset.

Table 11 summarizes the results of the proposed method and the existing methods which are discussed in Sect. 2 (related work).

Table 11 shows that the proposed methodology gives better accuracy i.e., 94.37% when compared to other algorithms.

Table 11 Evaluation summary

Author's name	Algorithm utilized	Precision (%)
Zhang et al.	Sparse autoencoder	75.27
Li et al.	Autoencoder, CNN	90.31
Afif et al.	DCNN	73.19
Roy et al.	NNELILS	94.00
Shahid et al.	Microsoft azure custom vision	90.00
The proposed method	DAECNN	94.37

6.4 Statistical analysis

McNemar's statistical test [24] is used to assess the efficiency of the algorithms CNN vs DAECNN, RBFMM vs DAECNN, and MLP vs DAECNN. The McNemar test is based on the chi-squared distribution using Eq. (2),

$$\chi^2 = \frac{(a - b)^2}{a + b} \quad (16)$$

where a is the number of instances misclassified by Classifier1 but not Classifier2 and similarly b denotes the count of instances that misclassified by Classifier2 but not Classifier1. The test statistics have a Chi-Squared distribution with 1 degree of freedom. At a 95% level of confidence, the difference of accuracies between the proposed and the existing methods CNN vs DAECNN, RBFMM vs DAECNN, and MLP vs DAECNN are significant as 1.0, 0.98, 1.0 which is greater than 0.5 respectively. Consequently, the null hypothesis can be rejected and the alternative hypothesis that states the classifiers have a different proportion of errors on the test set can be accepted.

7 Conclusion and future work

In this study, the deep neural network approach is employed to determine the current position of the visually impaired individual in a closed environment. The DAECNN algorithm combines two different neural networks such as denoising autoencoder and convolutional neural networks to efficiently classify the users' present location. Denoising autoencoder is applied to reconstruct the image from the noise input which helps to improve the classification performance. The advantage of the DAECNN approach is (1) Reconstruct the noisy image using denoising autoencoder that helps to improve the classification performance (2) CNN model automatically extracts the prominent features without any human intervention. The experimental results show that the proposed method efficiently classifies the images when compared to the existing approaches. Detect challenges and choose to navigate through a deep learning approach such as the recurrent neural network (RNN), Faster RCNN, and others for visually disabled individuals in the future.

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Declarations

Conflict of interest The authors have no conflict of interest.

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